

ENAE 641: Linear System Dynamics

Introduction:

This course is a practical introduction to the theory of multivariable, linear dynamical systems and feedback control design. The material is a fundamental building block for a graduate program in aerospace control systems.

Topical Outline:

(in rough chronological order)

1. Linearization of nonlinear systems; linear state space models
2. Fundamental solutions, state transition matrix, general solution
3. Linear vector spaces; linear algebra review
4. LTI matrix exponential; eigenstructure analysis; modal decompositions
5. Stability of linear systems; Lyapunov stability criteria
6. Input/output stability
7. Controllability and Observability; Gramians
8. Coordinate transforms and canonical forms
9. LTI singular value decompositions; multivariable frequency response
10. Full state feedback techniques; pole placement
11. Linear Quadratic Regulators (optimal state feedback)
12. State Observers
13. Observer/Controller designs; separation principle
14. Canonical servo loop and MIMO compensators
15. Tracking; disturbance and noise rejection
16. The Continuous Kalman Filter; LQG compensators
17. “Loop shaping” augmentation/tuning for LTI performance and robustness
18. Introduction to H_∞ design

Optional/recommended textbook: Hespanha, J., *Linear Systems Theory, 2nd ed.*.